ChatGPT

Winter Outdoor Risk for Dogs: Evidence, Public Data, and a Deterministic Local-First Model

Executive summary

This research treats winter dog-walking risk as a **measurement and inference problem**, not as an app-design problem. The governing brief is to determine what can be inferred from dog phenotype, latitude/longitude, time, and free/public environmental data while remaining deterministic, explainable, conservative, computationally light, and suitable for mostly local computation. filecite turn0file0

The central finding is that a useful system is feasible, but it should be **much narrower than the typical "winter safety score."** The strongest buildable signals are: **cold-exposure conditions; freezing-precipitation/ice conditions; observed or forecast precipitation and wetness potential; darkness/low visibility; and, optionally, measured particulate pollution.** Road/pavement temperature and road-surface state become unusually valuable wherever public Road Weather Information System (RWIS) data are available. FHWA describes RWIS stations as measuring variables such as pavement temperature, pavement freezing point, wet/icy state and sometimes pavement chemical concentration; NOAA MADIS can ingest corresponding state-DOT observations. [1]

The science does **not** support universal dog-specific temperature cutoffs of the sort commonly circulated online. Controlled and observational canine studies support body mass and coat morphology as thermoregulatory factors, and outdoor acclimatization can increase insulation and resting heat production. Veterinary guidance also identifies small size, thin coat, wetness, prolonged exposure, and exposed ears/feet/tails as relevant vulnerabilities. But the literature does not provide a validated function such as "a 9-kg short-haired dog is safe for 17 minutes at −8°C." [2] Accordingly, personalization should usually be **categorical and monotonic**—for example, "small + short/thin coat increases concern"—rather than a collection of invented numerical multipliers.

Human **wind chill should not be repurposed as a canine injury threshold**. The National Weather Service wind-chill index is explicitly based on heat loss from exposed human facial skin. Wind is nevertheless a physically meaningful canine input because it increases convective heat loss; the correct implementation is therefore to use wind directly as a cold-stress factor, not to tell an owner that a dog's "felt temperature" is the NWS human wind chill. [3]

Wet-cold exposure deserves disproportionately high attention. Veterinary guidance reports greater frostbite/hypothermia concern in wet and windy conditions, and canine cold-water experiments demonstrate rapid core-temperature decline under severe immersion. Those immersion studies cannot establish normal walking thresholds, but they support the direction of effect: **wetness compromises the protection implied by dry coat insulation.** [4]

The strongest winter surface model is an **ice/refreeze model, not a paw-temperature-injury model**. FHWA road-weather science shows that pavement temperature can differ substantially from air

temperature because of solar radiation, subsurface heat, surface characteristics, shade, wind and recent conditions, and that RWIS exists specifically because air temperature alone is inadequate for road icing decisions. [5] A pavement-energy-balance model is technically possible, but without calibrated surface material, shade, moisture and thermal-history information it should remain experimental for sidewalk-scale dog guidance.

Likewise, **road deicer exposure is a real canine hazard, but deicer presence is usually not remotely knowable.** Sodium chloride and other chloride deicers can irritate paws and gastrointestinal tissues; sufficiently large salt ingestion can produce dangerous hypernatremia and neurologic signs. [6] But weather compatible with salting is not evidence that a particular sidewalk was treated. Plow GPS is not proof of material application. Even states with excellent public winter-road systems usually expose pavement condition or plow position more readily than per-segment chemical application. Wisconsin, for example, publishes a developer API for winter road conditions, while Iowa publicly shows plow activity and operates pavement/weather stations; neither example justifies turning a nearby plow into "salt present." [7] A safety-oriented system should say **"treatment status unknown"** rather than manufacture a salt probability.

The recommended system is therefore a set of **independent risk components**, not a weighted master formula:

Component	Recommendation	Basis
Cold exposure conditions	**Build**	Strong environmental data; moderate-to-strong directional canine physiology evidence. [8]
Ice / traction	**Build**	Freezing precipitation and freeze/refreeze can be detected or estimated; RWIS can provide direct road evidence. [9]
Wet-cold escalation	**Build**	Strong directional physiology/veterinary support; precipitation readily available. [4]
Darkness / low visibility	**Build**	Sunrise/sunset is locally computable; visibility/fog/ snow observations exist. [10]
Air quality / smoke	**Build as optional context**	$PM_{2.5}$ data are available, but canine dose-response thresholds are not sufficiently established for precise personalization. [11]
Snow-depth/exertion	**Experimental**	Physiological direction plausible and some working-dog evidence exists; sidewalk-level snow depth is poorly represented by gridded data. [12]
Road-treatment/deicer exposure	**Measured-only plug-in**	Real hazard; nationwide exposure data unavailable. [13]
Exact pavement/sidewalk temperature from weather alone	**Experimental**	Physically modelable but highly site dependent. [14]

Component	Recommendation	Basis
Paw cold-injury temperature threshold	**Reject for now**	No validated consumer-dog injury threshold was found; available cold-sensation and sled-paw studies do not establish one. [15]
Breed-specific “safe temperature” charts	**Reject**	Available evidence supports morphology more strongly than precise breed-specific cutoffs. [16]
Weather-inferred “salt probably present”	**Reject for safety decisions**	Treatment depends on jurisdiction, route, maintenance action and pavement state; weather is only a weak proxy. [17]

The best **medium scope** is therefore recommended: U.S.-first with Canada-compatible weather adapters; six independent condition families; strong reliance on NWS/ECCC plus optional RWIS/state DOT data; dog personalization limited to evidence-backed morphology; and explicit `unknown` states where environmental evidence is missing. This is substantially more defensible than trying to model every conceivable winter hazard.

Research framing, scope, questions, and evidence base

Three scopes are plausible.

Scope	What it includes	Advantages	Main weakness	Recommendation
Narrow	Cold + precipitation + freezing/ice only; U.S. NWS; size and coat modifiers	Extremely cheap, simple and nationally available	Misses darkness, visibility, air quality and the unusually useful RWIS opportunity	Viable prototype
Medium	Cold, wetness, ice/refreeze, measured road state where available, visibility/daylight, optional $PM_{2.5}$; U.S. plus Canadian weather compatibility	Captures the majority of reliably estimable winter conditions without pretending to know sidewalk chemistry or exact paw temperature	Requires source-quality and state-DOT adapters	**Recommended**

Scope	What it includes	Advantages	Main weakness	Recommendation
Broad	All above plus inferred surface temperature by material, snow exertion, deicer likelihood, breed-specific cold limits, local sidewalk conditions	Feature-rich	Several outputs would be much more precise than the evidence or data justify	Not recommended

The medium scope best matches the design criteria in the supplied research brief because it concentrates complexity where the information gain is real. filecite turn0file0 NOAA/NWS can provide national U.S. forecasts, observations and alerts without a conventional private API secret; Environment and Climate Change Canada's GeoMet provides anonymous, free access to Canadian weather/environmental products and explicitly supports mobile applications. [18] Transportation data can then improve specific locations opportunistically rather than becoming a hard national dependency.

The principal research questions and testable hypotheses are:

Question	Working hypothesis	Evidence status after review
Can winter thermal risk be personalized from morphology?	Mass and coat class should be useful; precise breed-level temperature adjustments will not be defensible.	Supported. [19]
Does wind add information beyond air temperature?	Yes physiologically, but human wind-chill values should not be treated as canine thresholds.	Supported. [20]
Is wetness a meaningful escalation signal?	Yes, particularly with cold and wind.	Supported directionally. [4]
Can ice risk be derived cheaply?	Freezing rain is a strong direct signal; wet + subsequent freezing provides a useful but less certain refreeze signal.	Supported by road-weather practice. [21]
Can sidewalk/pavement temperature be estimated from weather?	Yes in principle, but not accurately enough everywhere without site-specific state/material/shade/history.	Supported; deployment confidence limited. [14]
Can deicer exposure be inferred from weather?	No—not with the reliability appropriate to a safety claim.	Supported by the mismatch between treatment operations and available public feeds. [22]
Can snow depth support dog-specific exertion guidance?	Coarsely, especially in open/unplowed terrain; not reliably for a sidewalk from lat/lon alone.	Evidence/data limited. [23]

Question	Working hypothesis	Evidence status after review
Is AQI useful in winter?	Yes as a separate environmental exposure signal, but human AQI categories are not validated canine clinical thresholds.	Environmental case strong; canine calibration weak. [24]
Can everything run locally?	Computation can; current environmental measurements still normally require network calls, exposing at least a coarse location.	Supported by API designs. [25]

Source priority. The evidence hierarchy used here is intentionally conservative. Tier one consists of peer-reviewed canine physiology/veterinary studies, NOAA/NWS/NCEI, FHWA, EPA, USDA/NRCS, ECCC and state DOT primary documentation. Tier two includes veterinary-school clinical guidance where controlled studies are absent. Poison-control/veterinary toxicology guidance is used for deicers where direct experimental exposure studies are sparse. General pet blogs and commercial "cold weather charts" are excluded as evidence.

The most important veterinary literature is surprisingly small and heterogeneous. A controlled seasonal exercise study found body mass among the important variables in canine thermoregulation, with small dogs facing greater heat-loss consequences because of scaling; a later agility-dog study again identified body mass and coat morphology as relevant factors. [16] Older outdoor-acclimatization work demonstrated increased winter insulation and resting heat production, while another cold-acclimation experiment found substantially increased energy expenditure during prolonged cold acclimation. [26] These findings strongly support **directional personalization**, but their populations, protocols and outcomes are not a basis for consumer walk-duration thresholds.

A useful methodological point follows from canine metabolic scaling: metabolism does not simply scale linearly with body mass. A study of 117 dogs estimated a body-mass exponent around 0.885 for basal metabolism, reinforcing why size has physiological consequences but also why crude "small/medium/large" multipliers should not masquerade as an exact biophysical model. [27]

The evidence is weaker for some commonly proposed attributes. **Age** appears in veterinary guidance as a vulnerability factor for very young and older dogs, yet age was not a dominant determinant in at least one controlled thermoregulation model. [28] Age should therefore produce a conservative uncertainty/escalation flag rather than a mathematically precise thermal penalty. **Brachycephalic status** is well known to matter to heat dissipation, but this review did not find evidence sufficient to assign a quantitative winter cold penalty. Snout dimensions occur in thermoregulatory studies, but that is not equivalent to validation of a cold-specific brachycephalic adjustment. [29] **Fur color** likewise lacks evidence for a useful winter cold-tolerance correction; any solar-absorption effect is secondary to coat structure, radiation, posture and exposure geometry and should not be implemented as a cold score.

Winter hazard taxonomy and dog-specific vulnerability

The scientifically useful taxonomy separates **hazard existence** from **dog susceptibility**. Cold air may be measured accurately while the exact likelihood of hypothermia in a particular dog remains uncertain; an icy

road can be measured directly even though the owner has not stated whether the dog has osteoarthritis. Treating these as separate layers prevents uncertainty in one domain from contaminating another.

Winter hazard	Mechanism and evidence	Dog modifiers worth using	Evidence grade	Modeling position
General cold exposure / hypothermia	Prolonged cold can overwhelm thermoregulation; veterinary guidance identifies wetness, wind and poor cold adaptation as aggravating factors. [30]	Mass/size, coat density/type, wetness, acclimatization; age conservatively	A/B directionally; weak for exact thresholds	Recommended
Frostbite / exposed extremities	Ears, feet, tail and exposed skin have less protection; freezing conditions and wind increase concern. Actual tissue freezing requires sufficiently low tissue/ environmental temperatures; human WCT times are not canine thresholds. [31]	Thin coat/exposed skin; morphology; prolonged exposure	B	Recommended as freezing-exposure flag, not injury timer
Wet dog + cold	Water dramatically increases heat transfer; severe canine immersion experiments produce falling core temperature, while veterinary sources warn about wet/windy exposure. [32]	Coat state matters more than dry coat classification once saturated; small mass plausibly compounds loss	A for physical direction, C for ordinary-walk thresholds	Recommended escalation
Wind-driven convective loss	Wind increases convective heat transfer; NWS WCT itself is human-specific. [20]	Smaller body size and less insulation plausibly increase sensitivity	A physically; B canine personalization	Use wind directly

Winter hazard	Mechanism and evidence	Dog modifiers worth using	Evidence grade	Modeling position
Cold paw contact	Canine paws are adapted to contact conditions, but research does not supply a validated pavement-temperature injury function. Healthy-dog cold sensory testing measured avoidance responses rather than tissue injury. [33]	Paw condition, coat/interdigital hair may matter; not all are available inputs	C/D for numerical injury model	Reject precise threshold
Ice / slick-surface slipping	Poor traction is clinically problematic, particularly for dogs with orthopedic/ neurologic impairment; gait studies document behavioral/kinetic changes on slick surfaces. [34]	Known mobility impairment strongest; age is only an imperfect proxy	B	Recommended environmental hazard
Freezing rain / glaze / black ice	Supercooled precipitation freezes on surfaces; thin clear ice can be difficult to see. Road temperature and road state are better evidence than air temperature alone. [35]	Mobility limitation, small/unstable gait; do not infer disease solely from age	A environmental / B dog impact	Recommended
Refreeze after melt/ slush	Liquid water on a surface followed by surface cooling below freezing can refreeze; surface temperature depends on solar, ground heat, wind and exposure. [36]	Mobility as above	A mechanism / B–C remote inference	Recommended with confidence label

Winter hazard	Mechanism and evidence	Dog modifiers worth using	Evidence grade	Modeling position
Deep/loose snow exertion	Working dogs expend substantial energy in winter travel; one hunting-dog study in 20–40 cm loose snow suggested higher locomotion cost than travel on prepared tracks. [37]	Small stature relative to snow depth, conditioning/ activity level	B mechanism / C generalization	Experimental
Wet snow / snowballs between toes	Veterinary guidance recognizes interdigital snow/ice accumulation, particularly with longer hair; sled-dog paw work found paw stress varied with snow conditions. [38]	Long/interdigital hair	B/C	Context warning, not numeric score
Packed/ crusted snow	Changes footing and paw loading; sled-dog evidence cannot be generalized cleanly to household dogs. [39]	Paw/orthopedic status	C	Experimental
Road/ sidewalk deicers	NaCl, $CaCl_2$, $MgCl_2$ and KCl are commonly encountered; contact can irritate paws/skin and ingestion from licking can cause gastrointestinal effects; sufficiently large salt ingestion can cause severe hypernatremia. [6]	Paw injury/skin disease and renal disease would matter, but are outside current profile	A for toxicity at sufficient dose; C for remote exposure	Measured-only

Winter hazard	Mechanism and evidence	Dog modifiers worth using	Evidence grade	Modeling position
Sand/grit mixtures	Primarily a mechanical/irritant concern rather than a systemic toxin; composition varies by jurisdiction. State winter operations often combine abrasive and chemical strategies depending on temperature. [40]	Paw condition	C	No quantitative model
Darkness / twilight	Darkness changes owner/vehicle visibility. Solar position is deterministic from location/date/time; NOAA's published method is based on standard astronomical algorithms. [41]	Fur color can affect *visual conspicuity*, but route/lighting/ reflective equipment dominate and are not known	A environmental; weak dog-specific	Recommended context
Snow/fog visibility	ASOS observations include visibility and present weather; snow and fog can materially reduce visibility. [42]	No meaningful physiological personalization	A environmental	Recommended context
Winter $PM_{2.5}$ / wood smoke / inversions	Wood burning can contribute materially to winter $PM_{2.5}$; smoke particles affect respiratory/ cardiovascular systems, while canine experimental/ observational evidence demonstrates plausible effects but not a validated pet-walking threshold. [43]	Known cardiopulmonary disease would matter but is absent from supplied profile; brachycephaly alone should not be numerically adjusted without evidence	A pollutant measurement / B–C canine threshold	Optional independent signal

Winter hazard	Mechanism and evidence	Dog modifiers worth using	Evidence grade	Modeling position
Antifreeze/ ethylene glycol puddles	Veterinary sources recognize antifreeze ingestion as a winter toxicologic emergency. [44]	None useful from current profile	A hazard / D remote detection	Do not dynamically model

The resulting **personalization evidence matrix** is deliberately less aggressive than common consumer advice.

Dog characteristic	Defensible relationship	Strength	How a deterministic model should use it
Body mass / size	Smaller dogs have less favorable size-related heat balance; mass repeatedly appears in thermoregulation studies. [16]	**A/B**	Categorical cold-vulnerability escalation
Coat density / undercoat	More insulating fur reduces dry heat loss; outdoor winter acclimatization can increase coat insulation. [45]	**A/B**	Thin/single vs dense/ double categories; avoid breed folklore
Coat length	Correlates imperfectly with insulation; density/undercoat can be more important; long hair also promotes snow accumulation around feet. [46]	**B**	Secondary modifier
Wet coat	Water substantially increases heat loss and compromises dry-fur protection. [32]	**A/B**	Strong escalation; never let “double coat” cancel wetness
Age: puppy/senior	Veterinary guidance recommends added caution, but controlled thermoregulation evidence is less clear. [28]	**B/C**	Conservative one-direction flag; no numeric factor
Body condition	Mechanistically relevant through insulation/energy reserves, but the reviewed consumer-applicable evidence is insufficient for a precise correction	**C**	Consider adding body-condition score in future; initially uncertainty only
Breed	Breed captures multiple anatomical features but is a poor substitute when size and coat are known	**C**	Use to populate phenotype defaults only, requiring user confirmation
Breed group	Broad taxonomic/functional grouping does not establish a thermal threshold	**D**	Do not score directly

Dog characteristic	Defensible relationship	Strength	How a deterministic model should use it
Brachycephaly / snout length	Insufficient cold-specific evidence for a numerical winter penalty	**C/D**	Store, but do not modify cold score yet
Activity / conditioning	Muscular work produces heat, but deep snow raises energetic demand; fatigue and post-exercise wetness complicate the relationship. [47]	**B/C**	Use for snow-exertion context, not as “active dogs are safe in colder air”
Fur color	No defensible cold-tolerance correction identified	**D**	Ignore for thermal risk; potentially relevant only to visibility
Outdoor cold acclimatization	Outdoor-acclimatized dogs can increase insulation/resting heat production. [48]	**B**	Potential future modifier; never a safety override

A notable evidence gap is **planned duration**. Cumulative cold risk cannot be assessed from date/time alone. The current input specification should therefore be extended with intended outdoor duration—or the system must honestly label its cold output as an **instantaneous-condition assessment** rather than an exposure assessment. Hypothermia and frostbite are inherently time-dependent, while the available research does not justify silently assuming a standard walk duration. [44]

Environmental data, free APIs, and the U.S. road-weather landscape

For the U.S., the best foundational source is **NOAA/National Weather Service**, supplemented rather than replaced by specialized datasets. NWS API information is intended as open data and is available without usage fees; it requires an identifying `User-Agent`, applies reasonable anti-abuse rate limiting, and supports forecasts, observations and alerts. NWS also cautions that internet delivery is not guaranteed, an important point for a safety-oriented client. [49]

NCEI's newly current **Global Historical Climatology Network hourly (GHCNh)** is particularly valuable for validation. It replaced the legacy Integrated Surface Dataset and provides harmonized hourly/synoptic station observations including temperature, dew point, humidity, precipitation, wind, visibility, present weather and—in some source records—snow information. It is updated daily and designed for historical/bulk access rather than a low-latency walk-time request. [50]

For snow, **SNODAS/National Snow Analyses** supplies daily gridded estimates of snow depth and related snowpack variables over the continental U.S. at approximately 1-km spatial resolution, assimilating model, remote-sensing and ground observations. That is excellent for regional snow context and retrospective validation but too coarse to tell whether a particular city sidewalk has been cleared. [51] USDA's **SNOTEL** network provides free hourly snow-depth, snow-water, temperature and related observations, primarily in the mountainous western U.S. and Alaska; it is excellent validation data in its footprint but not an urban national snow-depth API. [52]

For Canada, ECCC's **MSC GeoMet** is unusually favorable to a local-first implementation: access is anonymous and free, uses open geospatial standards, includes real-time and archived weather/environmental products, and is explicitly documented as suitable for mobile applications. The Canadian high-resolution deterministic forecast system provides roughly 2.5-km fields over most of Canada, including temperature, precipitation, cloud, wind and humidity, while recent surface observations are exposed through GeoMet's OGC services. [53]

The major candidate sources compare as follows.

Source	Useful signals	Cost/ authentication	Coverage/ resolution	Latency/ history/ forecast	Mobile/local-first assessment
NWS API	Temp, dewpoint, apparent temp, wind/ gust, precip forecasts, snow forecast, observations, alerts	Free; identifying User-Agent; no conventional private key. [54]	U.S.; gridded forecast + station observations	Current + forecast; observations can lag operational ingestion; cacheable	**Best U.S. core source.** Native HTTPS calls are straightforward; coordinate lookup necessarily reveals a location to NWS
ASOS/ METAR via NWS/ MADIS	Temp, dew point, wind, visibility, fog, precipitation/ present weather including freezing precipitation	Public observations. [55]	Point stations, airport-heavy	Hourly plus special observations; some higher-frequency data	Excellent measured-condition complement; station representativeness must be carried as uncertainty
NCEI GHCNh	Historical temperature, precipitation, wind, visibility, weather type	Free bulk/ search access. [56]	Global fixed stations	Daily archive update; long history; no forecast	**Validation/ research**, not walk-time primary
NWS alerts	Winter storm, freezing rain/ ice-related official warnings/ advisories	Open web service. [57]	U.S. polygons/ zones	Near-real-time active alerts	Lightweight high-value contextual input

Source	Useful signals	Cost/ authentication	Coverage/ resolution	Latency/ history/ forecast	Mobile/local-first assessment
NOHRSC/ SNODAS	Snow depth, snowpack temperature, SWE, melt	Free public gridded data. [51]	CONUS; ~1 km, daily	2003-present archive; current daily analysis	Useful cached regional layer; payload/access format is heavier than a simple mobile weather call
USDA SNOTEL	Snow depth, SWE, temp, soil variables	Free. [52]	Western U.S./Alaska points	Hourly observations; substantial history	Validation/special terrain source, not national urban dependency
NOAA solar calculation	Solar elevation, sunrise/ sunset	Offline calculation possible	Global from lat/lon/time	Fully deterministic	**Ideal local computation.** NOAA notes its online calculator is no longer actively maintained, but documents the underlying astronomical method and expected accuracy. [58]
EPA AirNow	Current AQI/ $PM_{2.5}$ and forecasts	API account/ key; observations are preliminary. [59]	U.S. plus reporting in Canada/ Mexico	Observations typically hourly; forecast products	Useful, but an embedded API key is operationally unattractive; optional rather than core
EPA AQS	Validated historical pollutant observations	Public API/ data	U.S. regulatory monitors	Validation data may lag substantially. [60]	Excellent retrospective AQ validation

Source	Useful signals	Cost/ authentication	Coverage/ resolution	Latency/ history/ forecast	Mobile/local-first assessment
ECCC MSC GeoMet	Canadian observations, forecasts, snow/ weather/ environmental layers, AQHI products	Anonymous and free. [61]	Canada plus some global products; HRDPS ~2.5 km. [62]	Real-time/ archive/ forecast depending collection	**Best Canadian core.** No private key; OGC complexity and response sizing require careful queries
ECCC NAPS/ AQHI	$PM_{2.5}$ and other pollutants/ AQHI	Government open data; GeoMet AQHI layers available. [63]	Canadian station/ network coverage	Hourly monitoring for continuous NAPS species	Strong Canadian air-quality option
MADIS RWIS	Air + pavement temperature, pavement state/freeze point, sometimes chemical concentration, visibility, precip	NOAA service/ API is free to meteorological users; web access may require account and datasets have provider restrictions. [64]	Participating U.S. DOTs; some Canadian coverage. [65]	Providers range from ~5 min upward	**High information value, medium integration cost.** Best as enhancement, not national hard dependency
FHWA WxDE	Fixed/mobile transportation weather observations and QC	Mostly freely available; registration/ restrictions for some datasets. [66]	Contributor-dependent	Near-real-time subscriptions + archive	Research/ validation valuable; FHWA has described it as a research platform rather than the ideal permanent operational dependency. [67]
State 511/ RWIS feeds	Road condition, pavement sensor data, cameras, sometimes plows	Varies by state	State/route specific	Usually near-real time	High-value plug-ins; no nationwide uniform API

Transportation data is the highest-value non-weather opportunity and the least standardized. FHWA's RWIS definition encompasses exactly the variables needed to improve an icing model: air temperature/humidity, precipitation, pavement temperature, pavement freeze point, wet/icy state, chemical concentration and subsurface conditions. [68] FHWA also notes that nighttime pavement temperature may vary by 10°F or more across a network because of exposure, altitude, traffic and materials—one reason point pavement observations should not be blindly extrapolated to a distant sidewalk. [69]

There **is** a partial national aggregation pathway, but not a universal, clean consumer road-treatment API. NOAA MADIS supports RWIS/Clarus variables, common formatting and QC, while FHWA WxDE collects fixed and mobile transportation weather from contributing agencies. [70] Availability and redistribution restrictions still depend on the provider, and state systems expose different subsets. Thus "national standardized RWIS data exists" would be too strong; a more accurate description is **federally supported aggregation of heterogeneous participating systems**.

State examples illustrate the variation:

Jurisdiction/source	Publicly useful information	What it does *not* prove
Minnesota DOT RWIS	MnDOT reports a network of roadside sites providing near-real-time surface and atmospheric information, and its public data server exposes current weather-sensor files. [71]	A nearby sidewalk has been salted
Utah DOT	Roughly 200 state weather stations measure variables including temperature, humidity, precipitation, soil temperature, snowfall rate, visibility and road conditions. [72]	Every local road/sidewalk shares the sensor's condition
Iowa DOT	Public plow positions; roadside weather stations with air/pavement/subsurface variables; explicit salt/brine winter operations. [22]	A displayed plow necessarily applied chemical at the user's location
Wisconsin 511	A documented API returns segment-level winter road condition/surface information; developer key and throttling are required. [73]	Chemical species, concentration or sidewalk state
Washington DOT internal winter telemetry example	Internal dashboard metadata includes road temperature and granular application-rate concepts. [74]	The telemetry is public—the dashboard explicitly says those vehicle data are internal

This heterogeneity argues for an **adapter architecture**:

`national atmospheric source → optional state road-weather adapter → optional municipality adapter`

rather than making the core model dependent on discovering a salt/plow endpoint in every city.

Road **treatment itself should carry a strict evidence hierarchy**:

`measured pavement chemical concentration` → strongest;

`explicit agency treatment/application event for the segment` → strong;

`documented recent plow with spreader/application telemetry` → moderate-to-strong if the telemetry explicitly contains rate/material;

`plow location only` → weak;

`weather conditions under which a jurisdiction commonly treats roads` → weak proxy;

`generic statement that winter roads are often salted` → not a location-specific measurement.

Only the first three should drive a dynamic chemical-exposure warning.

Derived-signal feasibility and the buildability matrix

A useful deterministic system needs to label every result by epistemic type. The following vocabulary is recommended:

Measured means the relevant environmental variable was directly observed by an authoritative sensor/feed. **Derived—high confidence** means the relationship follows from well-established physics using adequately measured inputs. **Derived—moderate confidence** means the mechanism is sound but spatial/surface uncertainty is important. **Weak proxy** means correlation exists but the signal cannot reliably establish the local hazard. **Not reliably knowable** means available remote data do not answer the question.

The complete buildability assessment is:

Hazard	Dog impact	Dog modifiers	Environmental inputs	Best source	Status	Scientific confidence	Data availabi
Cold ambient exposure	Hypothermia/ thermal stress	Mass, coat, age cautiously, acclimation	Air temp, wind, planned duration	NWS/ECCC	Measured inputs + derived condition	High direction / moderate calibration	Exceller

Hazard	Dog impact	Dog modifiers	Environmental inputs	Best source	Status	Scientific confidence	Data availabi
Wet-cold exposure	Faster heat loss	Coat state, mass	Active rain/ sleet/wet snow; user “dog wet” state would improve model	NWS/ ECCC/ ASOS	Derived—high/ moderate	High direction	Excellen
Freezing-exposed-tissue conditions	Frostbite concern	Exposed ears/tail/ feet; coat	Air ≤ freezing, wind, duration	NWS/ECCC	Derived—moderate	Moderate	Excellen
Wind chill as canine “feels like”	Potentially misleading	—	Human WCT	NWS formula	Wrong biological target	Low	Excellen
Active freezing rain	Glaze/traction	Mobility vulnerability	Precipitation type, official alert	ASOS/ NWS/ECCC	Measured/ forecast	High	Good
Existing icy roadway	Slip/fall	Mobility limitation	Road surface state	RWIS/511	Measured	High	Patchy l
Refreeze	Slip/fall	Mobility limitation	Recent liquid precip/melt + surface temp; otherwise air-temp history	RWIS + NWS	High confidence with pavement temp; moderate with air temp	High mechanism	Mixed
Black ice exact sidewalk	Slip/fall	Mobility limitation	Microlocal surface temp/ water/shade	None national	Weak proxy	Mechanism high; local detection low	Poor
Pavement/ sidewalk temperature	Paw exposure / ice state	Limited	Air/history, solar, cloud, wind, material, moisture, ground temp	RWIS ground truth; weather inputs	Derived—moderate at best without calibration	High physics; low transferability	Mixed

Hazard	Dog impact	Dog modifiers	Environmental inputs	Best source	Status	Scientific confidence	Data availabi
Paw cold-injury threshold	Tissue injury	Paw condition	Surface temp + contact time	No sufficient source	Not reliably calibrated	Low	Poor
Snow depth	Exertion/ impedance	Size/ stature, activity	Snow depth	SNODAS/ SNOTEL	Measured/ model analysis, coarse	Moderate	Good regiona
Snow on sidewalk	Exertion/paws	Size, coat	Plowing/local accumulation	No national source	Weak proxy	Moderate	Poor
Interdigital ice balls	Paw discomfort	Long foot hair	Wet snow + temp	Weather + phenotype	Derived—moderate	Moderate	Good environ inputs
Deicer chemical present	Paw/GI chemical exposure	Paw/skin condition	Direct chemical/ applicator record	RWIS/ application telemetry	Measured where available	High hazard	Poor/ heterog
Deicer inferred from snow/ cold	Potential false reassurance/ alarm	—	Weather/ policy	Weather + policy	Weak proxy	Low as exposure detector	Easy
Darkness	Human/ vehicle visibility	Fur color only weakly relevant	Solar position	Local calculation	Derived—high	Very high	Univers
Fog/snow visibility	Human/ vehicle visibility	None	Visibility, present weather	ASOS/ ECCC	Measured	High	Good n stations
$PM_{2.5}$ / smoke	Respiratory/ cardiovascular exposure	Health state would be more relevant than supplied phenotype	$PM_{2.5}$/AQI	AirNow/ ECCC	Measured	High environmental / moderate canine guidance	Good populat areas
Antifreeze puddle	Severe poisoning if ingested	None	Puddle identity	None	Not reliably knowable	Hazard high/ detection absent	None

The key derived signals deserve additional scrutiny.

Freeze/thaw and refreeze. With a measured pavement temperature, surface wetness/precipitation and pavement condition, a deterministic freezing-state rule can be quite strong because those are precisely the variables RWIS systems are designed to collect. [75] Without pavement temperature, "recent wet conditions + air temperature crossing below 0°C" remains useful as a **refreeze-possible** indicator but not proof of ice; pavement and air temperatures can diverge substantially because of sun and ground heat. Iowa DOT explicitly warns that pavement and air can differ by as much as roughly 20°F in some circumstances. [76]

A conservative rule therefore looks like:

```
if observed_road_state == ICY:
    ice_state = MEASURED_HIGH

elif observed_freezing_rain:
    ice_state = HIGH_CONFIDENCE_AREA_HAZARD

elif pavement_temp <= 0°C and surface_wet:
    ice_state = HIGH_CONFIDENCE_AT_SENSOR

elif recent_liquid_precipitation and air_temperature crossed from >0°C to <=0°C:
    ice_state = REFREEZE_POSSIBLE
    confidence = MODERATE

else:
    ice_state = UNKNOWN_OR_LOW_EVIDENCE
```

"Unknown or low evidence" must not be rendered as "no ice."

Surface temperature. A physical pavement model would solve an energy balance involving incoming solar/long-wave radiation, convective exchange, conduction into the substrate, latent heat/moisture and surface optical/thermal properties. FHWA materials show that operational pavement prediction uses atmospheric conditions plus pavement/subsurface information, while observed temperatures vary materially with exposure and construction. [77] Consequently, a single formula based only on air temperature, wind and sunshine should not drive a paw-injury decision. Its legitimate use is as a research model validated against RWIS and stratified by surface class.

Snow depth/exertion. SNODAS can describe regional ground snow depth daily, and working-dog studies confirm that snow travel can impose substantial energy costs. [78] But an urban lat/lon does not establish whether the walking surface is an unplowed field, freshly shoveled sidewalk, packed trail, or salted pavement. A future system could make snow-depth personalization defensible only after adding **route/surface context** or a direct user observation.

Deicer likelihood. Weather can establish that agencies *might* deploy treatment, and agency policy can describe treatment practices. Iowa, for example, explains when salt/brine are used and how pavement temperature controls deicing effectiveness. [17] That still cannot prove the user's sidewalk was treated. For a safety-oriented product, "policy says crews may pre-treat before this storm" is informative metadata, not chemical detection.

The engineering ranking is therefore:

Priority class	Signals
High value + high confidence + easy data	Air temperature; wind; active precipitation/type; freezing rain; recent temperature history; wet-cold flag; darkness; official winter alerts
High value + moderate confidence	Refreeze from recent wetness + temperature trajectory; dog cold vulnerability from mass/coat; low visibility; RWIS ice/pavement temperature where locally available
Promising but data-limited	Snow-depth exertion; direct deicer exposure; road-surface temperature; pavement-condition interpolation
Interesting but not currently reliable	Sidewalk material-specific temperature; crusted-snow paw risk; neighborhood black-ice localization from atmospheric data alone
Do not build	Breed-specific safe-temperature charts; human wind chill converted to dog "feels like"; exact frostbite countdowns; weather-inferred salt presence; exact paw injury thresholds; remotely inferred antifreeze-puddle risk

Minimum viable deterministic model and local-first architecture

The minimum viable model should expose **independent risk facts with independent confidence**, because averaging unrelated hazards produces unsafe semantics. A severely icy surface does not become "moderate" because air quality is excellent, and missing salt data does not become "low chemical risk" because temperature is known.

A sensible first version has five core components plus one optional component.

Component	Inputs	Transformation	Dog-specific modifiers	Output	Confidence/ failure rules	Freshness
Cold Exposure Conditions	Air temp, wind, planned duration, precipitation/ wetness	Deterministic condition flags; monotonic duration escalation; do not use human WCT as dog threshold	Mass/size and coat strongest; age only conservative; wetness can override coat benefit	`normal / cold-stress conditions / elevated-vulnerability conditions`, plus reasons	No duration → "instantaneous only." Missing wind → lower confidence. No breed-specific degrees	Current obs ideally ≤1 h; forecast valid a walk time

Component	Inputs	Transformation	Dog-specific modifiers	Output	Confidence/ failure rules	Freshness
Wet-Cold Escalation	Rain/sleet/wet snow, recent immersion/ user-wet flag if available, temp/wind	Explicit wetness flag combined with cold exposure	Coat state, size	`wet-cold concern` with reason	Weather gives *potential to get wet*, not proof dog is wet; user confirmation materially improves it	Precip observation/ forecast contemporane
Ice / Traction	Freezing rain, recent precipitation, temp trajectory; RWIS pavement temp/state when available	Evidence hierarchy: measured icy state > freezing rain > wet pavement ≤0°C > air-based refreeze proxy	Mobility limitation if explicitly known; age alone only weak modifier	`measured ice / high-confidence icing / refreeze possible / unknown`	No “clear” state from air temp alone	RWIS/road condition preferably tens minutes; atmospheric ≤
Visibility / Darkness	Lat/lon/time, solar position; measured visibility/fog/ snow	Solar calculation offline; measured visibility threshold bands based on transport visibility rather than canine physiology	None initially	`daylight / twilight-darkness`; `visibility reduced`	Route/traffic not known, so describe environmental visibility rather than collision probability	Solar timeless; visibility ≤1 h
Winter Chemical Exposure	Only direct application event, chemical concentration, or explicit treatment feed	No weather-based substitution	None from current phenotype	`treatment confirmed / treatment data unavailable`	**Unknown is mandatory default**	Treatment reco must be event-current

Component	Inputs	Transformation	Dog-specific modifiers	Output	Confidence/ failure rules	Freshness
Air Quality *(optional)*	$PM_{2.5}$/AQI	Retain official environmental category/ concentration rather than creating canine cutoffs	No phenotype adjustment initially; health conditions would be needed	`particle pollution elevated`	Human AQI is environmental context, not validated canine dose-response	Hourly/current

Snow depth should remain an **experimental seventh module**, disabled from strong decision-making until route context is available.

Dog personalization should be modeled as an evidence-qualified **vulnerability profile**, not a breed lookup table:

```
thermal_vulnerability:
    body_size_class:        supported
    coat_insulation_class:  supported
    age_vulnerability:      cautious
    wet_coat:               strong escalation
    acclimatization:        optional / moderate evidence
    brachycephaly:           stored, no cold adjustment
    fur_color:               ignored for thermal scoring
    breed_group:             ignored unless it supplies a user-confirmed
phenotype default
```

This also avoids a common ethical problem: **using breed as a stereotype when the actual phenotype is already known.** A whippet-like thin coat or a dense double coat is more directly relevant than a kennel-club group label; the research literature supports morphology and mass more clearly than group-specific safe temperatures. [16]

Uncertainty should be first-class data. Every environmental signal should carry at least:

```
value
valid_time
observed_or_forecast
source
measurement_class = measured | derived_high | derived_moderate | weak_proxy
spatial_representativeness
freshness
```

```
quality_flags
confidence
```

Then apply four deterministic safety rules:

1. **Unknown never becomes safe.** Missing pavement state produces `ice status unknown`, not `no ice`.
2. **More direct evidence dominates proxies.** An RWIS `ICY` observation outranks an atmospheric model that would otherwise predict no refreeze. FHWA's RWIS design explicitly exists to observe road-specific conditions that general weather can miss. [79]
3. **Low-confidence personalization never creates precision.** Age or breed may add explanatory caution but cannot transform a weak environmental signal into a precise danger threshold.
4. **Independent risks are not numerically averaged.** The overall outdoor state should be the most consequential supported component plus unresolved critical unknowns.

Conceptually:

```
overall_state =
    most_severe_supported(
        cold,
        wet_cold,
        ice_traction,
        visibility,
        chemical_if_measured,
        air_quality_optional
    )

also_return:
    critical_unknowns
    reasons
    source_times
    confidence_by_component
```

This is much safer than a 0–100 “winter score,” whose weights would have no physiological interpretation.

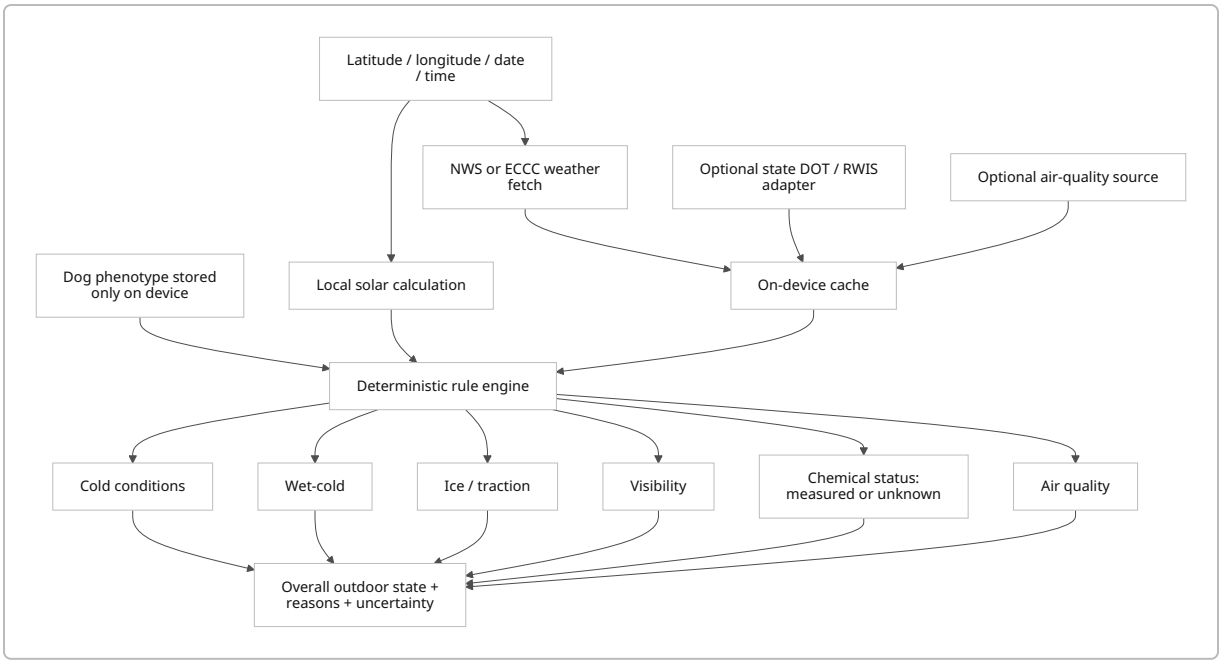
Local-first feasibility is strong for computation but conditional for acquisition. Solar calculations, dog profile storage, historical temperature-window processing, freeze/thaw logic, rule evaluation and all personalization can live entirely on-device. NOAA's published solar approach is deterministic from position and time. [58]

For U.S. weather, a native mobile client can call NWS HTTPS endpoints directly without embedding a conventional private API key; caching and polite request behavior remain necessary. [54] For Canada, GeoMet access is anonymous and free and explicitly supports application integration, though OGC requests require more parsing than the NWS point API. [61] Native mobile applications are not subject to browser CORS in the way browser JavaScript is; a web implementation would need to validate each source's CORS policy separately.

Location privacy cannot be absolute while querying a coordinate-specific service. Any direct request containing lat/lon reveals that query coordinate to the service. A defensible privacy strategy is to keep the dog profile entirely local, quantize coordinates to no finer precision than the environmental source warrants, cache location-to-grid mappings, minimize refresh frequency and avoid transmitting a persistent user identifier. NWS forecasts are gridded, so sending unnecessarily precise GPS coordinates creates privacy cost without corresponding forecast resolution benefit. The exact quantization should be empirically tested at forecast-zone/grid boundaries rather than hard-coded blindly.

API-key requirements alter the architecture. AirNow requires a key/account and publishes preliminary observations, so putting an unrestricted production credential in a distributable binary is undesirable; air quality is therefore a good optional module for a tiny proxy/backend or an alternative public-file ingestion path, not a reason to introduce a backend for the entire system. [59] State 511 systems likewise vary: Wisconsin requires a developer key and throttles API use, while other public state files may be directly accessible. [80]

A lightweight data-flow architecture is consequently:



No generative model is necessary anywhere in this chain.

Validation, evidence gaps, research roadmap, risks, and dependency graph

A deterministic system is only trustworthy if its **derived variables are separately validated**. The validation target should not be “does the final score feel right?” Each intermediate prediction should be tested against an observable quantity.

Model/ component	Prediction to validate	Ground truth	Evaluation method	Important edge cases
Air-temperature/ wind acquisition	Value at correct location/time	ASOS/MADIS/ NCEI GHCNh	Source agreement, age/freshness checks	Mountain terrain, coastal gradients, station outages
Refreeze proxy	Surface becomes or remains frozen/ icy	RWIS pavement temperature + pavement condition; state 511 road reports	Classification calibration, false negatives/positives stratified by event	Bridges, shaded roads, post-salt pavement, strong solar heating
Pavement-temperature experimental model	Surface temperature trajectory	RWIS pavement sensors	MAE/bias plus prediction intervals, stratified by surface/site/ lead time	Bridges, snow-covered pavement, shaded cuts, urban heat islands
Freezing-rain detection	Freezing precipitation occurring	ASOS present-weather + official reports	Precision/recall by event	Mixed precipitation, sparse stations
Snow-depth context	Gridded snow depth	SNOTEL/COOP/ manual observations where available	Error by region/ elevation/season	Urban plowing, drifting, forest cover
Visibility	Reported horizontal visibility	ASOS	Error/freshness and station-distance sensitivity	Localized fog, snow squalls
$PM_{2.5}$	Current exposure estimate	AirNow initially; later validated AQS/NAPS archive	Bias/freshness and monitor-distance analysis	Wildfire plumes, inversions, sparse rural monitors
Dog cold personalization	Relative thermal response by phenotype	**No adequate public population ground truth found**	Prospective non-invasive veterinary study required	Small thin-coated dogs, elderly dogs, wet coat, highly acclimatized dogs
Paw-temperature/ injury model	Tissue temperature/ response during surface contact	**Dedicated experiment required**	Non-injurious thermal/contact measurements	Ice, saline slush, wet pads, snow

Model/ component	Prediction to validate	Ground truth	Evaluation method	Important edge cases
Deicer exposure	Chemical is actually present at walking surface	Maintenance logs plus surface chloride sampling	Detection precision/recall	Sidewalk vs roadway, private property, post-melt residue

An important validation principle is **not to predeclare an arbitrary universal "acceptable error."** The acceptable error depends on the decision boundary. For pavement freezing, for example, a model with a small mean absolute error can still be unsafe if its uncertainty interval repeatedly crosses 0°C and the implementation calls the road "nonfreezing." The release criterion should therefore be framed around the resulting decision: uncertainty intervals, calibration, and especially false-negative behavior around the relevant physical transition.

For derived pavement temperature, validation should be geographically blocked rather than randomly shuffled. Training/calibrating on one sensor and testing adjacent timestamps from the same sensor would overstate generalization; leave entire stations, road classes, climate regions and storms out for testing. FHWA's documented spatial variation in pavement temperature makes this particularly important. [69] Seasonal validation should explicitly include early winter when ground heat remains relatively high, midwinter deep-cold periods, spring thaw, sunny clear days, cloudy nights, freezing rain and rapid post-storm clearing.

Dog-specific validation is the largest scientific gap. The existing thermoregulation literature supports directional effects of mass, coat and acclimatization but does not validate consumer-facing cold-risk categories across the enormous variation in domestic dogs. [19] A future study should collect non-invasive skin/core-proxy temperatures, activity, coat characterization, mass/body condition, wetness and environmental conditions during ordinary voluntary walks, with predefined stop criteria and veterinary oversight. It should **not** experimentally induce frostbite, clinically significant hypothermia or painful paw injury. Healthy-dog cold-stimulation research demonstrates that non-injurious quantitative cold-response protocols are possible, but also shows habituation and methodological uncertainty. [33]

A corresponding paw study would need instrumented surface temperature, pad temperature, moisture, contact time/pressure and surface chemistry. The sled-dog literature is informative but not sufficient because highly conditioned working dogs on snow are not representative of the general pet population. [39]

For deicer inference, the needed experiment is primarily **environmental rather than animal-based**: obtain road/sidewalk treatment records where available, collect geolocated surface chloride/conductivity measurements after storms, and test whether weather, treatment policy, plow telemetry and time-since-storm can predict actual residues. Until such validation exists, the system should refuse to turn those variables into a chemical-presence claim.

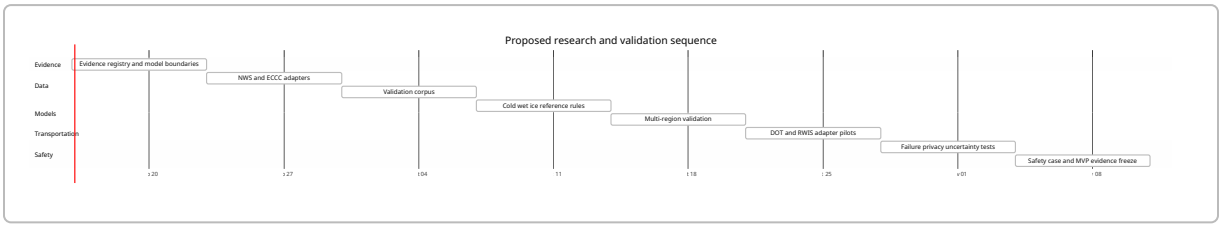
The prioritized engineering/research backlog follows directly from the evidence:

Rank	Work item	Why now	Release status
P0	NWS + ECCC core weather adapters with provenance/freshness	Foundational, authoritative, cheap/free	Production
P0	Offline solar/daylight calculation	Universal, deterministic, no network/privacy cost	Production
P0	Environmental freezing-rain/refreeze rules	High safety value and testable against public road sensors	Production after validation
P0	Dog vulnerability representation based on mass + coat, with cautious age flag	Best-supported personalization available	Production, categorical only
P0	Explicit `unknown/stale/proxy/measured` semantics	Prevents absence of data from becoming reassurance	Production
P1	State-DOT/RWIS adapter interface	Large incremental value for ice/pavement state	Production framework
P1	First high-quality state adapters, selected by public data quality rather than population alone	Demonstrates measured road-state enhancement	Production per state
P1	Visibility/fog integration	Low cost, independent safety value	Production
P1	Optional $PM_{2.5}$ module	Meaningful exposure signal; separate auth architecture	Optional production
P2	SNODAS snow-depth context	Useful for open terrain but poor sidewalk specificity	Experimental
P2	Pavement-temperature model calibrated against RWIS	Scientifically testable, potentially useful where sensors absent	Experimental
P2	Direct treatment/chemical adapters where an agency exposes real application telemetry	Strong signal but rare	Experimental/ local
P3	Prospective canine thermoregulation validation	Necessary before more precise dog-specific risk bands	Research
P3	Paw-contact laboratory/field validation	Necessary before any temperature/contact injury model	Research
Reject	Breed “safe temperature” table	False precision	Do not build

Rank	Work item	Why now	Release status
Reject	Human wind chill → canine frostbite countdown	Wrong reference organism/ model	Do not build
Reject	Salt probability from weather/plow proximity alone	Exposure not established	Do not build
Reject	Exact black-ice map from ordinary weather forecast alone	Microlocal state not observable	Do not build

An eight-week research-to-validation program would look like this; it is a proposed project sequence, not an assertion that every experimental evidence gap can be closed in eight weeks.

Phase	Milestone	Deliverable
Week 1	Evidence and variable freeze	Versioned evidence registry; allowed/forbidden personalization rules
Week 2	Core U.S./Canada acquisition	NWS and ECCC adapters; data provenance/freshness schema
Week 3	Historical validation corpus	NCEI weather + selected RWIS/511 station-event dataset
Week 4	Cold/wet/ice rules	Deterministic reference implementation with test vectors
Week 5	Ice/refreeze validation	Error/calibration report across multiple climates and events
Week 6	State transportation pilot	Two or more heterogeneous DOT adapters proving plug-in design
Week 7	Uncertainty and failure testing	Stale/missing/conflicting-data tests; privacy/network/cache tests
Week 8	Safety review and MVP freeze	Documented MVP rule set, rejected signals, known limitations and experiment backlog



The principal **risks and limitations** are scientific rather than computational. First, veterinary evidence is insufficient for precise individualized cold thresholds; deterministic formulas would not solve that evidence problem. [16] Second, weather fields describe an area, not the dog's exact microclimate. Third, road sensors describe particular road surfaces, not automatically a nearby sidewalk. FHWA explicitly documents

substantial pavement-temperature variation within road networks. [69] Fourth, municipal treatment practices and private sidewalk treatment can differ radically within a few meters, making deicer exposure intrinsically difficult to remote-sense.

There is also an asymmetry of error. A false warning generally costs inconvenience; a false "safe" assessment can contribute to injury. That justifies conservative semantics but **not** indiscriminately inflating every risk. The appropriate conservative behavior is to identify the uncertainty—"surface treatment unknown," "ice possible because only air-temperature data are available"—rather than converting every unknown into a high-risk alarm.

Animal-welfare ethics strongly favor observational and non-injurious validation. Experiments should not seek the environmental temperature at which dogs develop frostbite or clinical hypothermia. Existing studies involving severe immersion or induced hypothermia demonstrate biological mechanisms but should not be repeated simply to derive consumer thresholds. [81]

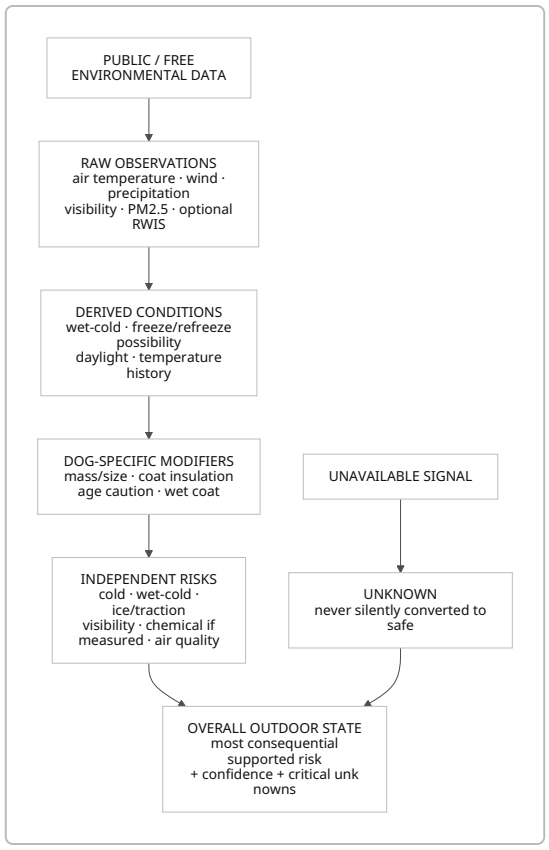
Breed-based personalization also deserves ethical restraint. Because within-breed variation and mixed ancestry are large and directly measured phenotype is more relevant to heat exchange, breed should be used, at most, to suggest defaults such as likely coat structure or body-size range; the owner should confirm the physical trait and the actual confirmed trait should drive the rule.

Finally, every output needs traceable provenance. A deterministic safety system should be able to explain, for example:

> "Ice/refreeze concern is elevated because precipitation was observed within the recent window and temperature crossed below freezing; no pavement sensor is available, so this is a moderate-confidence area-level inference."

That is much more defensible than either "Risk score 74" or a generative explanation unsupported by the underlying measurements.

The evidence ultimately supports the following dependency graph—and, importantly, several arrows intentionally terminate in `UNKNOWN` when the evidence is absent:



The smallest scientifically defensible answer to **“What winter conditions matter for this particular dog before going outside?”** is therefore not an all-knowing winter score. It is an evidence-ranked combination of **measured cold, wind and precipitation; wet-cold escalation; freezing/ice evidence; mass and coat-based vulnerability; daylight/visibility; optional air pollution; and direct road-treatment information only when it actually exists**. Everything else should either remain explicitly experimental or remain unknown until better measurements and canine validation data exist.

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